

CERYDA

A Market Validation Proposal for Decentralized Compute Storage and AI Infrastructure

Concept Paper v0.1

EARLY CONCEPT | PROPOSED FOR SOLANA | TOKEN NOT LAUNCHED

Purpose: public feedback and market validation

Document Status

Status	Early Concept
Network	Proposed for Solana
Token	Not yet launched
Purpose	Public feedback and market validation

This paper describes an early-stage proposal. It is not an offer to sell tokens, a solicitation of investment, or a promise that the network will be built. The purpose is to test whether users, infrastructure providers, developers, and ecosystem participants believe the concept solves a real problem.

1 Why We Are Publishing This

We are not launching a token today. We are not conducting a token sale. We are not asking anyone to invest money.

We are publishing this concept because we believe there may be an opportunity to build a new type of decentralized infrastructure network connecting computing power, storage capacity, AI workloads, infrastructure providers, developers, and users through an open digital marketplace.

Before committing significant capital or building a complex protocol, we want to answer a more fundamental question: Does the market actually want this?

This document describes what we are considering building, why the problem matters, how the network could function, and where a digital token may eventually fit within the ecosystem. We would rather build with feedback from potential users than spend years developing something nobody needs.

If you operate computing infrastructure, build AI applications, need access to compute or storage, develop on Solana, invest in infrastructure, or are interested in decentralized technology, we want to hear what you think.

2 The Problem

The digital economy is becoming increasingly dependent on physical computing infrastructure. Artificial intelligence requires enormous amounts of GPU compute, CPU compute, data storage, bandwidth, electrical power, and data-center capacity.

At the same time, computing resources exist throughout the world that may be underutilized. These resources range from individual workstations and GPU systems to professional servers, small data centers, specialized compute facilities, and large-scale infrastructure.

The problem is not simply that computing power does not exist. Access to computing infrastructure remains fragmented. A developer may need GPU capacity. A small AI company may need temporary compute without purchasing expensive hardware. A research team may require storage or processing capacity for a limited period. A business or data-center operator may own capacity that remains underutilized.

There is potentially value on both sides of that equation: one party needs compute, while another party has compute. The same concept applies to storage and other infrastructure resources. Ceryda would seek to connect those participants.

3 What We Are Proposing

Ceryda is a proposed decentralized infrastructure marketplace designed to connect resource providers with resource users.

Resource providers could contribute GPU compute, CPU compute, storage, server capacity, specialized AI hardware, and potentially other infrastructure resources in the future.

Resource users could seek infrastructure for artificial intelligence, machine learning, model inference, model training, rendering, scientific computing, data processing, application hosting, distributed storage, and other computational workloads.

The long-term goal is straightforward: make useful infrastructure easier to discover, access, verify, and compensate. Rather than requiring every participant to negotiate an individual contract with every provider, Ceryda would attempt to create a standardized marketplace.

4 A Simple Example

Imagine a small company developing an AI application. It needs access to high-performance GPUs but is not ready to spend tens of thousands of dollars purchasing its own hardware. Through Ceryda, the company could potentially submit a compute requirement, specify the hardware characteristics it needs, receive available capacity from qualified providers, pay for the workload, run the job, receive proof of completion, and evaluate the provider.

On the opposite side, imagine an infrastructure provider operating several GPU workstations that sit partially idle overnight. That provider could register eligible hardware, make part of its capacity discoverable, accept compatible workloads, perform the computation, provide verification of completion, and receive compensation.

Ceryda would act as the coordination layer between the two.

5 Why Blockchain

Ceryda does not begin with the assumption that every part of computing infrastructure belongs on a blockchain. Most computing would occur off-chain. The blockchain would instead serve as a coordination and settlement layer.

Potential on-chain uses include provider registration, payment settlement, escrow, network incentives, reputation records, staking, governance, workload-verification references, resource-market transactions, and treasury transparency.

The blockchain does not perform the AI workload. The infrastructure performs the workload. The blockchain helps coordinate the economic relationship surrounding that workload.

6 Why Solana

Ceryda is being explored with Solana as its blockchain settlement layer. The reasoning is practical. An infrastructure marketplace could eventually require a large number of relatively small interactions among customers, providers, applications, automated agents, infrastructure nodes, and treasury systems.

For that model to function efficiently, settlement needs to be fast and inexpensive. The architecture would therefore attempt to keep computation off-chain while using Solana where transparent digital settlement provides an advantage.

The final technical architecture has not been determined. That is one reason Ceryda is presented as a concept rather than a finished protocol.

7 The Provider Network

A major component of Ceryda would be the Provider Network. Providers could range from relatively small operators to professional infrastructure companies.

Potential provider classes include personal compute providers contributing qualified workstation or server capacity; professional compute providers operating dedicated GPU or CPU systems; data-center providers offering larger blocks of capacity; storage providers; and specialized infrastructure providers serving particular workloads.

Not every computer would automatically qualify. Hardware capability, network performance, reliability, security, uptime, and workload requirements would need to be evaluated. A provider-reputation system could help customers distinguish between different levels of service.

8 The User Side

For users, the experience should ultimately feel less like interacting with cryptocurrency and more like purchasing infrastructure. A user should be able to describe a workload, required hardware, time horizon, budget, data needs, and performance expectations without needing to understand every part of the blockchain layer.

Ceryda would seek to match that request with suitable capacity, communicate pricing and service terms, coordinate payment, and provide useful evidence that the requested work was completed.

9 Verification Trust and Reputation

A decentralized marketplace is useful only if participants can trust the claims being made. Providers may claim specific hardware, uptime, bandwidth, geography, security controls, or workload completion. Ceryda would need mechanisms for testing and recording those claims.

Possible approaches include hardware benchmarking, signed telemetry, challenge workloads, independent verification nodes, cryptographic attestations, customer feedback, dispute records, and provider-performance histories.

No single mechanism is assumed to solve every trust problem. The verification design would need to balance rigor, privacy, cost, speed, and resistance to manipulation.

10 The Possible Role of a Token

A token should not exist merely because the project uses blockchain. If Ceryda eventually introduces a digital token, it should perform necessary functions within the network.

Potential functions could include payment, provider collateral, staking, network incentives, governance, dispute participation, rewards for verification activity, and access to certain services. Some functions may be better handled with stablecoins or conventional payments.

No token supply, allocation, sale structure, ticker, or launch date has been established. Ceryda should not launch a token unless the network can demonstrate a credible need for one.

11 Payments Fees and Revenue

The marketplace could support payment in stable digital assets, fiat-linked payment methods, a future Ceryda network token, or a combination of these approaches. The final model should minimize friction for users.

A sustainable network may charge a transparent protocol or marketplace fee on completed transactions. Additional revenue could eventually come from enterprise services, APIs, verification services, premium provider tools, or specialized infrastructure coordination.

Any fee model would need to remain competitive with centralized and decentralized infrastructure alternatives while funding operations, security, support, development, and ecosystem growth.

12 Security Privacy and Workload Protection

Infrastructure marketplaces create real security risks. Workloads may contain sensitive data, providers may operate untrusted machines, and malicious users may submit abusive or illegal jobs.

Ceryda would need controls for workload isolation, encryption, access management, software integrity, provider screening, monitoring, incident response, data retention, and dispute handling. Different workload classes may require different security and compliance standards.

The protocol should avoid making claims of security, privacy, or regulatory compliance that it cannot verify.

13 Governance and Network Rules

If Ceryda becomes decentralized, participants will need clear rules for provider eligibility, verification standards, disputes, protocol changes, fees, treasury use, and emergency responses.

Governance should be introduced only where it improves accountability and resilience. Early development may require a more centralized operating structure, with decentralization expanding only as the network, tooling, and participant base mature.

14 What Has Not Been Decided

This concept intentionally leaves major questions open. The final architecture, legal structure, payment system, token design, provider requirements, verification mechanisms, governance process, fee schedule, launch network, and operating model have not been finalized.

Those decisions should follow evidence from users and providers, technical testing, security review, economic analysis, and legal advice.

15 What We Need to Validate

Before committing significant capital, Ceryda needs evidence that the market values the proposed network. The first validation questions are whether customers will use a new source of compute or storage, whether providers will list capacity, which workloads are most suitable, what verification customers require, what pricing is competitive, and whether blockchain adds enough value to justify its complexity.

The project should treat interest, interviews, waitlist signups, pilot commitments, technical contributions, and paid test workloads as stronger signals than social-media attention alone.

16 Early Participation

Ceryda is seeking feedback from compute and storage providers, AI and machine-learning developers, researchers, application builders, Solana developers, data-center operators, security specialists, infrastructure buyers, and people with experience building decentralized physical infrastructure networks.

Useful feedback includes objections. We want to understand why the concept would fail, which parts are unnecessary, what trust problems are hardest, and what would make participation worthwhile.

17 Proposed Early Roadmap

Phase 1: publish the concept, interview prospective users and providers, gather requirements, and test demand.

Phase 2: define a narrow pilot focused on one workload category and a small provider group.

Phase 3: prototype discovery, qualification, verification, payment, and reporting without introducing an unnecessary token.

Phase 4: run controlled testnet or off-chain pilots, measure reliability and economics, and revise the design.

Phase 5: decide whether a broader network, on-chain protocol, and token are justified by real usage.

18 Risks

Ceryda faces substantial risks, including insufficient market demand, competition from established cloud and decentralized-compute platforms, unreliable provider capacity, security breaches, verification failures, weak unit economics, regulatory uncertainty, token-design problems, and the difficulty of delivering a simple user experience across heterogeneous infrastructure.

Publishing this paper does not eliminate those risks. It is a low-cost way to expose them earlier.

19 Call for Participants

If you need compute or storage, operate underused infrastructure, build AI applications, work on verification systems, develop in the Solana ecosystem, or have experience with infrastructure marketplaces, we invite you to review this concept and share practical feedback.

The immediate objective is not fundraising or token promotion. It is to determine whether Ceryda should be built, which problem it should solve first, and who is willing to help test that solution.

20 Closing

The opportunity is simple to describe but difficult to execute: useful infrastructure exists, demand for infrastructure is growing, and the systems for connecting supply with demand remain fragmented.

Ceryda proposes an open coordination layer that could make decentralized compute, storage, and AI infrastructure easier to discover, verify, use, and compensate. The next step is to test that proposition with the people who would actually use and operate the network.